



LUND UNIVERSITY

Final Report

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1. Product concept

GreenGuard monitor is a small plug and play device designed for growers who want real time tracking over their farming of plants and mushrooms. It measures temperature, humidity, total VOC and equivalent CO₂. It then displays it on an alphanumeric display while a neopixel shifts to different colours if the condition is outside the optimal threshold and a buzzer starts beeping.

2. Technical requirements

The system is initialized by the microcontroller, first testing all I²C addresses (AHT20, SGP30, HT16K33) and creating the corresponding object. If any module is missing, an error is logged. The program then enters the main loop, organised as a simple state diagram. In state 1, CO₂, TVOC, temperature and humidity are read and the values are displayed on the alphanumeric display. In state 2, all measurements are compared against defined limits (TVOC $\geq$ 2000 ppb, CO₂ $\geq$ 1000 ppm, temp $>$ 32 °C, RH $>$ 70%). If no limit is exceeded, the system goes to state 3 for a 1s delay before returning to state 1 again. If any threshold is exceeded, the logic jumps to state 4, starts a buzzer signal, and the corresponding light on the Neopixel, depending on the type of alarm set and continues to display values. When the user presses the button, the buzzer stops, the PWM output is deactivated, and the alarm flag is reset. Then the loop returns to state 1 when the values return within the threshold again.

3. Description and analysis of the system

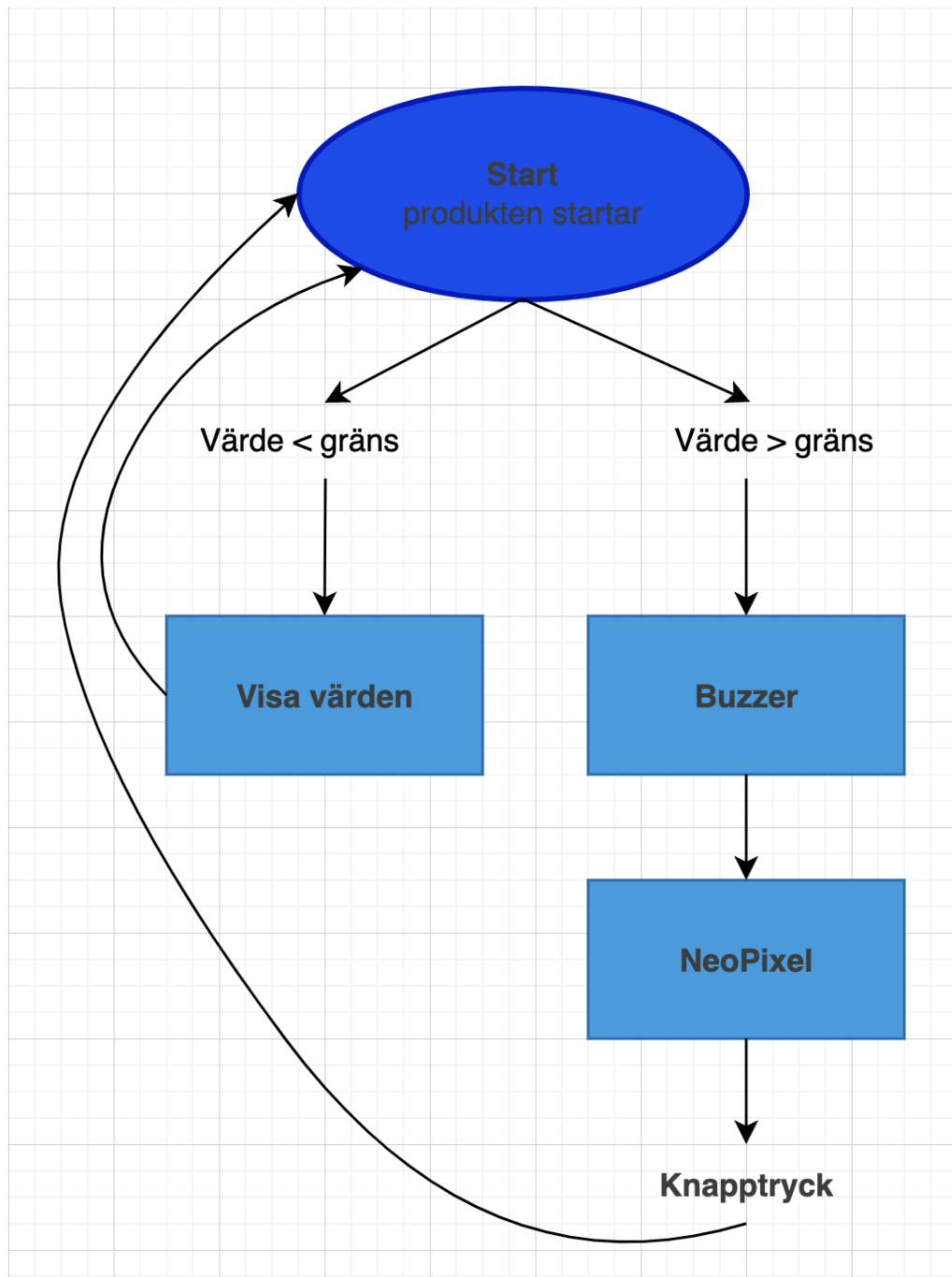
3.1 Hardware specifications

- Adafruit Feather RP2040
- SGP30 (CO2/eVOC)
- AHT20 (Temperature/Humidity)
- Piezo Buzzer
- Alpha numeric Display

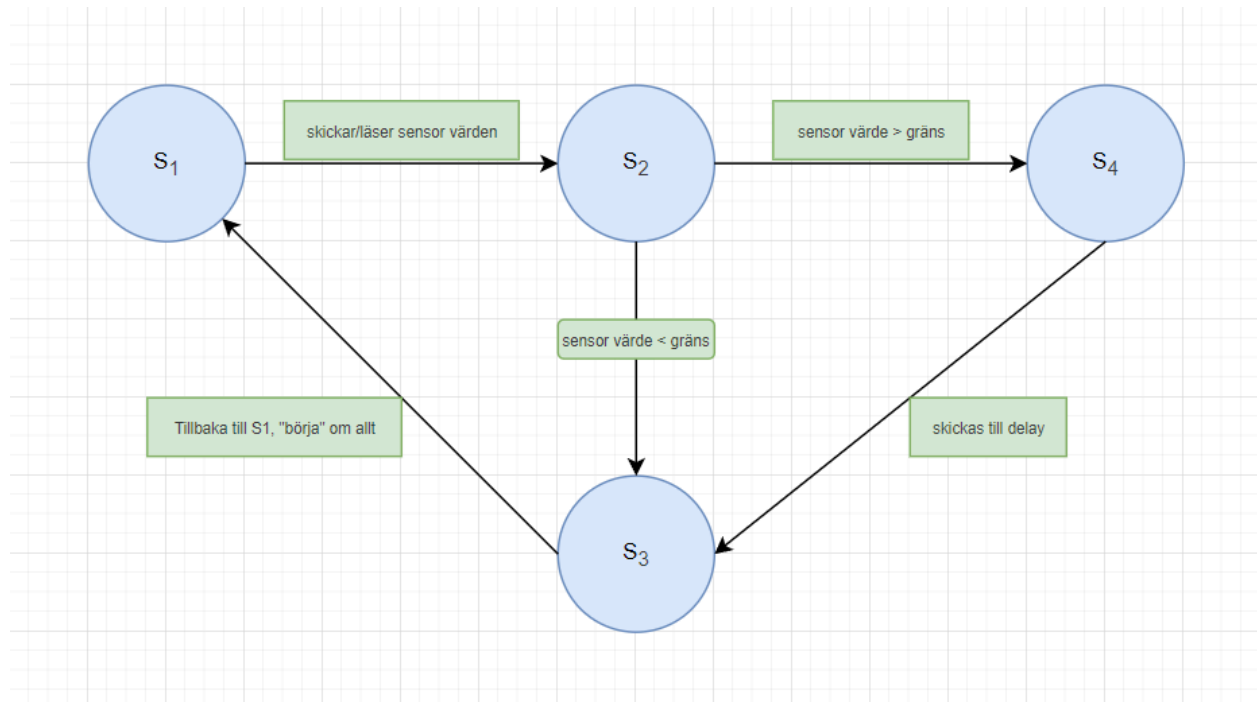
3.2 Software specifications

- Circuit python
- Libraries:
 - Adafruit circuitpython bundle
 - “adafruit_sgp30” for CO2/eVOC readings
 - “adafruit_ahtx0” for temperature/humidity data
 - “adafruit_ht16k33” for alphanumeric display control
 - “pwmio” for PWM control
 - “board” for access to adafruit IO ports
 - “neopixel” for neopixel led control
 - “busio” for I2C bus
 - “digitalio” for IO control

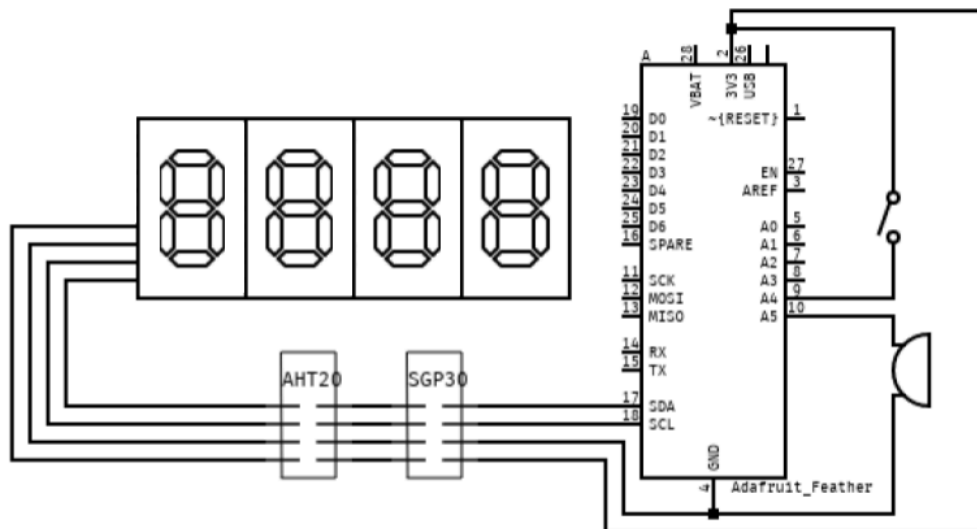
3.4 Block diagram



3.5 State diagram



3.6 Circuit diagram



3.7 Results and outcome

We believe that the expected outcome of the GreenGuard monitoring system has been achieved. The product fulfills our goals of easily being able to analyze values from the various sensors, which in turn are displayed on the screen for the user. The user can then monitor the environment and assess whether preemptive measures are in order. In addition, are there default thresholds that sound an alarm when the environment is at a dangerous level. The alarm is sounded through a piezo buzzer and a color is displayed on the neopixel, informing the user of what is wrong. To then stop the buzzer, the button needs to be pressed and the system restarts.

In conclusion we as a group believe that GreenGuard monitoring has fulfilled all our criteria and achieved our believed result.

4. Contribution statement

We split the work so each of us could focus on our strengths, but every decision was made together to keep quality high and collaboration smooth. Alan led the coding and made sure the software ran smoothly, with input from the rest of the team on important features. Samuel designed the circuit schematic, chose the right connection and tested the hardware. Anton created the state and block diagram, and also worked closely with Dlovan on the diagrams, furthermore he was the group leader and made sure everything was handed in on time, and ensured that the plan was being followed. Dlovan kept an eye on documentation and made sure everything was documented correctly, from the technical specification to the final report. Throughout the whole project we all helped each other and discussed every major decision as a group.

5. User guide

1. To start the device, plug in a power source either into a USB power or a 3.3 V supply.
2. Let the system initialize, and settle in the various sensors. After roughly 10 seconds the alpha numeric display shows the various sensor values.
3. Reading the display:

The Alpha numeric Display cycles through the values every 2 s:

- Temperature in C
- Humidity in %
- Total VOC in ppb
- Equivalent CO₂ in ppm,

4. Alarm behaviour

If and only if the buzzer starts, will the neopixel display different colours, below is the colour that refers to different sensor indicating thresholds:

- Red - Temperature over threshold
- Green - TVOC over threshold
- Blue - Humidity over threshold
- White - eCO₂

To turn off the alarm, press the button once.

5. Shutdown procedure

- Unplug the power supply at any time.